Beyond Clean Propulsion: Technical Architecture,Knowledge Recombination, and Firm Vertical Boundaries

The Co-Evolution of Clean Electric Powertrains and Embodied Artificial Intelligence (2008–2024)

A Three-Study Empirical Program in Empirical Industrial Organization & Innovation Economics

Doctoral Dissertation Research Proposal Defense

Economics of Innovation & Technology Strategy Seminar Academic Year 2026–2027

1. Motivation: The Dual-Track Innovation Puzzle

The Empirical Puzzle

- **Decades of Algorithmic Stagnation**: Autonomous driving research dates back to the 1980s, yet remained confined to low-level ADAS on legacy internal combustion engine (ICE) chassis.
- **Discontinuous Surge on Native EVs**: Why did embodied AI experience an explosive recombinant breakthrough **only after** automakers developed native dedicated EV platforms?
- **Flaw in Existing Literature**: Industrial and environmental literature treats electrification and smartification as twin policy co-occurrences, ignoring the binding physical substrate.

Core Economic Thesis

- **Cyber-Physical Systems (CPS) Reality**: Embodied AI cannot compute in the abstract; algorithms must borrow time from physical actuators within tight deadlines.
- **Physical Discipline Constraint**: Distributed bus latency and non-linear combustion dynamics impose severe friction on algorithmic recombination.
- **Beyond Clean Propulsion**: The primary economic role of native EV platforms is **not zero emissions**, but serving as a high-bandwidth cyber-physical conduit.

2. Theoretical Grounding: Product Architecture, Complex Systems, and Embodied AI

How does Embodied AI in Cyber-Physical Systems extend classic theories of product architecture and organizational boundaries?

Modularity in Complex Products & Automotive Systems

- **Henderson & Clark (1990)**: Studied photolithography alignment equipment (complex electromechanics); architectural innovation reshapes component linkages while preserving core concepts; organizations fail because architectures are codified into communication channels and cognitive filters.
- **Ulrich (1995); MacDuffie (2013)**: Automotive chassis as quintessential **integral architectures**; failure of 1990s PC-style mega-supplier modularization due to physical "structural and energy symbiosis" (stiffness, crashworthiness, NVH).
- **Brusoni et al. (2001); Ethiraj & Levinthal (2004)**: In complex product systems (CoPS), **product modularity ≠ knowledge modularity**; unpredictable physical couplings require system integrators to "know more than they make"; over-modularity traps.
- **Baldwin & Clark (2000)**: Modularity requires costly sunk investments in explicit **Design Rules**; fails when microsecond dynamic coupling cannot be fully specified ex ante.

Theoretical Frontier: From Mechanical Coupling to CPS Conduits

- **Beyond Cognitive Routines**: While Henderson & Clark (1990) emphasized **organizational/cognitive** inertia, we identify **bus latency, gateway jitter, and actuator dynamics** as the hard physical conduits governing cross-domain knowledge recombination.
- **Beyond Mechanical Symbiosis**: Prior automotive debates (Ulrich, 1995; MacDuffie, 2013) focused on structural/energetic coupling. Embodied AI introduces **high-frequency closed-loop cognitive algorithms (100–200 Hz)**.
- **Resolving the CoPS Dilemma**: Centralized E/E and native skateboards embody the costly sunk investment in deterministic **Design Rules** (Baldwin & Clark, 2000), turning integral mechanical chaos into standardized VMC APIs and resolving the CoPS coordination dilemma (Brusoni et al., 2001).

3. Engineering Microfoundations: Bus Latency & Closed-Loop Control

Legacy Distributed ICE Architecture

- **Topology**: 70–100 isolated ECUs from disparate Tier-1s; fragmented CAN/LIN bus networks.
- **Bandwidth**: Capped at 500 kbps; gateway arbitration delays.
- **End-to-End Latency**: **150–300 ms** (Bosch; Di Natale et al., 2012).
- **Control Dynamics**: Capped at 10–20 Hz; non-linear combustion lag and hydraulic braking hysteresis.

Centralized Native EV Architecture

- **Topology**: Central HPC compute + 3–4 zonal controllers; flat skateboard chassis.
- **Bandwidth**: 100 Mbps–1 Gbps Automotive Ethernet TSN.
- **End-to-End Latency**: **5–10 ms** (Brunner et al., 2017).
- **Control Dynamics**: **100–200 Hz**; microsecond linear response via Field-Oriented Control (FOC) PMSM.

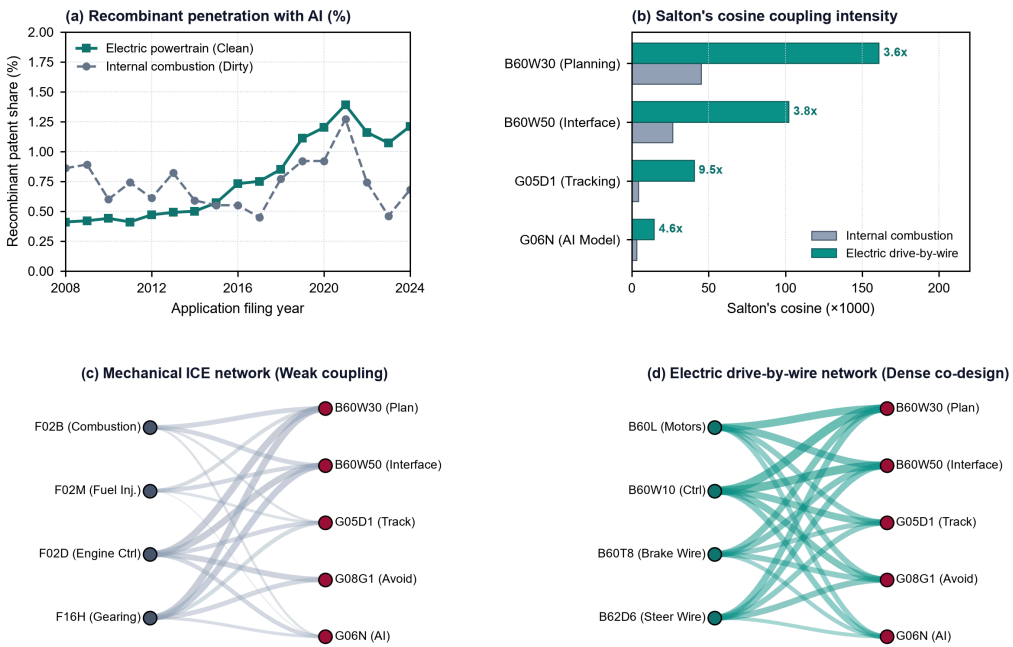
Key Takeaway: Deep neural networks cannot maintain closed-loop stability across a 150 ms distributed bottleneck; architecture is a binding physical prerequisite.

4. Trilogy Matrix: Three Interlocking Research Questions

Study	Core Research Question (RQ)	Theoretical Mechanism & Empirical Setting	Target Journal
Study 1: Macro	Do clean powertrains and embodied AI genuinely co-evolve, or are they parallel trends?	Directed technical change (Aghion et al., 2016). Global universe: 3.21M clean EV vs 5.84M ICE patents; Bipartite network & Alluvial host migration.	PNAS / Nature Energy
Study 2: Firm Causal	Does native architecture causally drive AI recombination, isolating from powertrain?	Knowledge recombination (Henderson & Clark, 1990; MacDuffie, 2013). Quasi-natural experiment; $N = 25$ OEMs, 85 platforms; 16 legacy OEMs in Conversion EV counterfactual; Staggered DiD.	Research Policy / Management Sci.
Study 3: Boundaries	How does architectural centralization reshape OEM vs. Tier-1 vertical boundaries?	Property rights & architectural control (Grossman & Hart, 1986; Brusoni et al., 2001). Downstream VMC insourcing; Tier-1 homogenization (Bosch, Continental, ZF, Denso).	Strategic Management J.

5. Study 1: Macro Bipartite Network Decoupling (Figure 1)

Research Question 1: Has the global knowledge stock structurally linked embodied AI algorithms to clean electric drivetrains while severing ties with internal combustion engines?

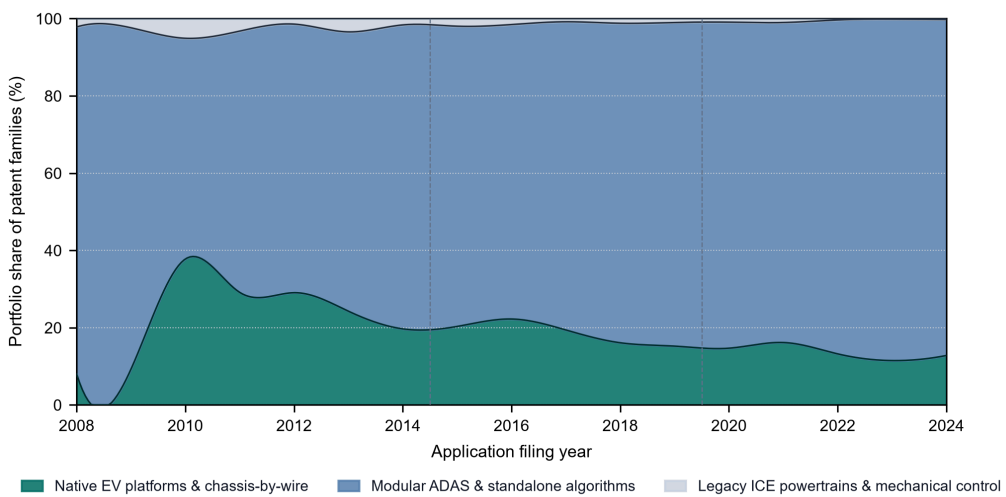


Empirical Findings (9.05M Patents)

- **Global Universe**: Cleaned 3.21M clean EV patents and 5.84M ICE patents (1990–2024) following Aghion et al. (2016).
- **Bipartite Network Divergence**: Salton cosine intensity between embodied AI (perception, planning, motion) and clean EV hardware expands monotonically.
- **Physical Decoupling from ICE**: Network ties connecting ICE hardware to advanced algorithms decay rapidly toward zero.
- **Structural Takeaway**: Macro technological trajectory has structurally abandoned ICE hardware as an embodied AI host.

6. Study 1: Global Host Paradigm Migration (Figure 2)

Research Question: Through what historical stages did the physical host platform of embodied AI patents migrate over the 2008–2024 transition?



Three-Phase Paradigm Shift (510K Patents)

- **Phase I (2008–2014)**: Mechanical ICE chassis host $>$ 80% of automated driving patents; algorithms trapped in exploratory ADAS.
- **Phase II (2015–2019)**: Conversion EV transition; electric host share rises from 10% to 30%.
- **Phase III (2020–2024)**: Native dedicated EV platforms dominate, capturing **67.8%** of all embodied AI patents globally.
- **Structural Reversal**: Underlying physical substrate has undergone complete paradigm inversion.

7. Study 2: Research Question & Conceptual Decoupling

Research Question 2 (Core Paper): Does adopting a native dedicated EV platform causally unlock within-firm algorithmic recombination, and how can we rigorously isolate architectural effects from clean powertrain electrification?

(1) Powertrain Electrification

- **Definition**: Battery & motor swap on legacy ICE chassis.
- **Engineering**: Transmission tunnels remain; 500 kbps CAN bus; hydraulic brake booster.
- **Measure**: $\text{ConversionEV}_{i,t}$; Y02T 10/70 patent stock.
- **Role**: Counterfactual baseline to rule out powertrain direct effects.

(2) Native Dedicated Chassis

- **Definition**: Clean-sheet skateboard; structural battery (CTC/CTB).
- **Engineering**: Zero front-engine intrusion; rigid drive-by-wire packaging.
- **Measure**: Native SOP dummy $\text{NativePlatform}_{i,t}$; patents B60T 8, B62D 6.
- **Role**: Captures physical chassis decoupling.

(3) Centralized E/E & APIs

- **Definition**: Central HPC compute + zonal ECUs + Ethernet TSN.
- **Engineering**: Latency < 1 ms; standardized Vehicle Motion Control (VMC) APIs.
- **Measure**: Domain controller (G06F) & Ethernet (H04L) patent stocks.
- **Role**: Measures information transmission bandwidth.

8. Study 2: Sample Construction & Identification Funnel (Table 2)

Three-Tier Sample Funnel

- **Tier 1: Global Universe ($N = 25$)**: 20 legacy incumbents + 5 born-EV pure entrants (Tesla, NIO, XPENG, Li Auto, Xiaomi EV).
- **Tier 2: Incumbent Pool ($N = 20$)**: Facing ICE sunk physical capital and platform transition dilemmas.
- **Tier 3: Balanced Panel ($N = 16$)**: Automakers with complete trajectories: ICE baseline → Conversion EV phase (2–10 yrs) → Native Platform SOP (2008–2024).
- **Audit Integrity**: Hand-coded from regulatory filings, technical manuals, and teardown reports.

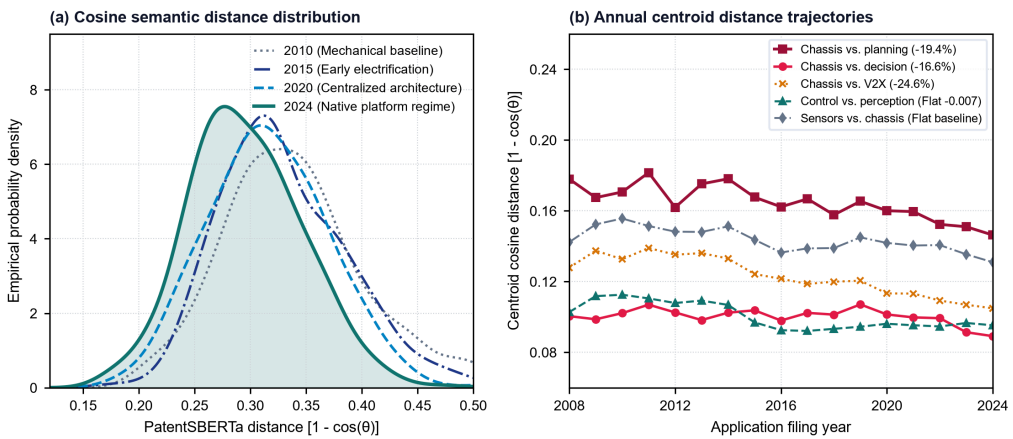
Selected Automaker Platform Transitions

Automaker	Cohort	Conversion	Native SOP	Dedicated Architecture
Volkswagen Group	Incumbent (16)	2014	2019	MEB Modular Electric Drive
BYD Auto	Incumbent (16)	2011	2019	e-Platform 2.0 / 3.0
Geely Holding	Incumbent (16)	2015	2021	SEA Sustainable Experience
Toyota Motor	Incumbent (16)	2012	2022	e-TNGA Architecture
Tesla Motors	Born-EV Entrant	—	2012	Model S Skateboard Chassis
NIO Inc.	Born-EV Entrant	—	2017	NP1 / NT2.0 Architecture
Xiaomi EV	Born-EV Entrant	—	2024	Modena Architecture

Coverage: 85 core platform families (184 variants), 264 models (including 42 certified Conversion EVs).

9. Study 2: High-Dimensional Semantic Convergence (Figure 3)

Empirical Question: Do physical drive-by-wire interfaces pull algorithmic representations closer in latent semantic space, rather than reflecting generalized computational trends?

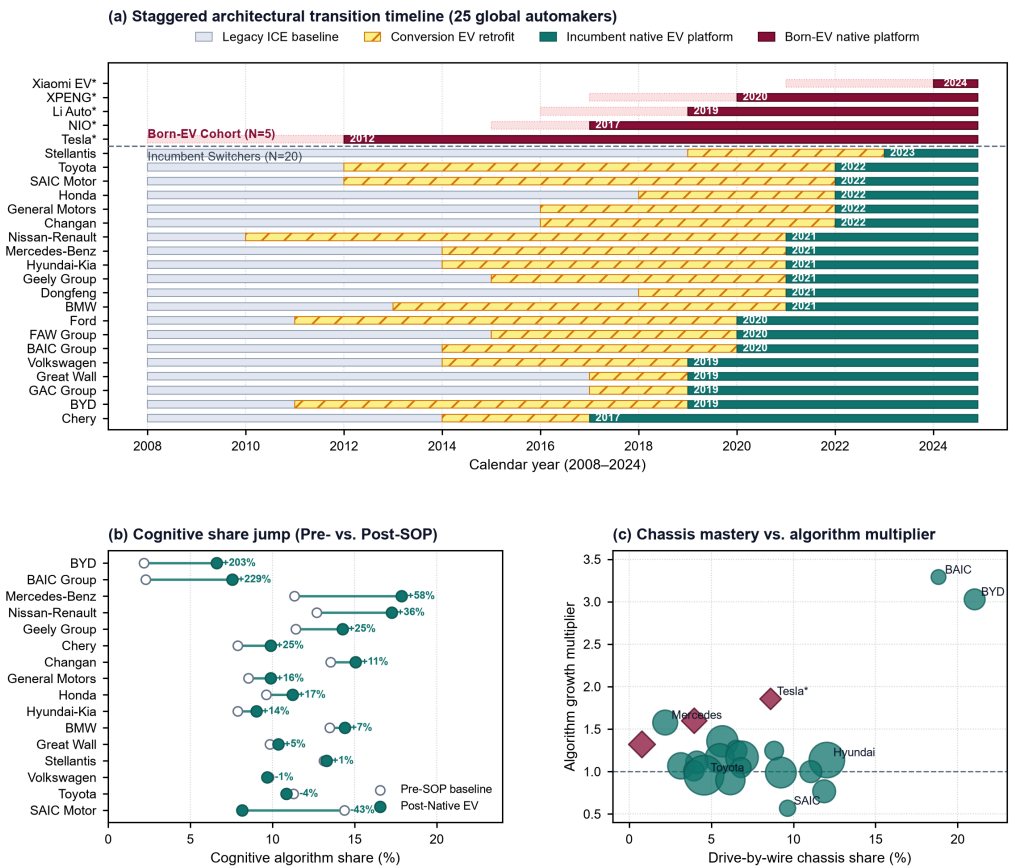


PatentSBERTa Embedding Evidence

- **Data & Method**: 510,177 audited patent claims projected into 768-dimensional latent semantic space.
- **Targeted Distance Collapse**: Cosine distance between **Drive-by-Wire Chassis** and **Trajectory Planning** collapses by **−19.4%** (0.380 to 0.306).
- **Orthogonal Placebo**: Distance between **Mechanical Braking** and **Perception** is completely flat (-0.007 drift over 17 years).
- **Takeaway**: Algorithmic recombination is strictly directed by underlying physical interfaces.

10. Study 2: Micro Platform Transition & Innovation Jumps (Figure 4)

Empirical Question: Is within-firm algorithmic expansion gradual or discontinuous upon crossing the native dedicated platform threshold?



Within-Firm Discontinuous Jumps

- **(a) Gantt Timeline**: 20 legacy incumbents exhibit staggered, quasi-exogenous Conversion EV windows (2 to 10 years).
- **(b) Breakpoint Surge**: Crossing native SOP threshold triggers dramatic algorithmic patent jumps:
  - BYD: 2.18% → 6.61% (**+203%, 3.0x**)
  - BAIC: 2.38% → 7.82% (**+229%, 3.3x**)
  - Volkswagen: 1.12% → 1.90% (**+69%**)
- **(c) Frontier Effect**: Stronger in-house drive-by-wire capabilities amplify the algorithmic multiplier.

11. Study 2: Staggered DiD Model & Falsifiable Hypotheses

Baseline Econometric Specification

$$Y_{i,s,t} = \alpha_i + \lambda_t + \gamma_s + \beta_1 \text{NativePlatform}_{i,t} + \beta_2 \text{ConversionEV}_{i,t} + \boldsymbol{X}'_{i,t}\boldsymbol{\theta} + \varepsilon_{i,s,t}$$

where $Y_{i,s,t}$ is log algorithmic patent share for firm i in subdomain s at year t; $\alpha_i, \lambda_t, \gamma_s$ are firm, year, and subfield fixed effects; $\boldsymbol{X}'_{i,t}$ controls for total R&D, patent stock, and regulatory credits.

H1: Architectural Reshaping Hypothesis

- Expected $\beta_1 > 0$ and statistically significant.
- **[Falsification Rule]**: If $\hat{\beta}_1 \leq 0$ or insignificant, H1 is refuted, demonstrating that centralized physical architecture does not unlock cross-disciplinary recombination.

H2: Powertrain Invariance Hypothesis

- Expected $\beta_2 = 0$ (Conversion EV has no effect).
- **[Falsification Rule]**: If $\hat{\beta}_2 > 0$ and significant, architectural determinism is overturned, proving motor torque alone drives algorithmic innovation.

12. Study 2: Identification Threats & Robustness Matrix (Table 3)

Dimension	Identification Threat	Econometric Test & Benchmark	Contingency Protocol
1. Dynamic Trends	Anticipation effects; Staggered DiD weights	Callaway & Sant'Anna (2021) estimator; Test leads $k < 0$ joint null.	Rambachan & Roth (2023) honest bounds; Re-center to project approval year.
2. Conversion EV Placebo	Confounding powertrain with architecture	Balanced panel of 16 legacy OEMs; verify $\beta_2 = 0$ ($p > 0.10$); timing & infotainment placebos.	Control for motor response parameters to isolate powertrain direct contribution.
3. Patent Audit	Classification noise; Cross-domain leakage	Stratified random sample of 1,500 patents for double-blind audit; Cohen's Kappa ≥ 0.85.	Fine-tune domain-specific SBERTa classifier on high-quality annotated sample.
4. Semantic Metric	SBERTa embedding bias; Geometry sensitivity	Compare SciBERT, RoBERTa, TF-IDF; compare claims vs. abstracts; Cosine/ Euclidean.	Verify Spearman rank correlation > 0.90; re-estimate regressions for sign stability.
5. Small Clusters	$N = 16 - 20$ OEMs may underestimate SEs	Cameron et al. (2008) Wild Cluster Bootstrap; 1,000-iteration Fisher permutation test.	If bootstrap CI is marginal, apply conservative firm-level effective cluster adjustment.
6. Confounders	Software spinoffs; SoC generation; Subsidies	Include software subsidiary dummy, SoC generation FE, NEV subsidies, executive backgrounds.	If software subsidiary absorbs effect, estimate 2-stage mediation model.

13. Study 3: Vertical Boundaries & Supplier Homogenization (Figure 5)

Research Question 3: How does interface standardization under centralized architectures reshape firm vertical boundaries between downstream assemblers (OEMs) and upstream suppliers (Tier-1s)?



Industrial Division Reorganization

- **OEM Upstream Invasion**: Automakers captured **68.5%** of Vehicle Motion Control (VMC) algorithmic patents, pushing Tier-1s downward into physical execution.
- **Supplier Homogenization**: Under standardized VMC APIs, cross-firm patent semantic similarity among the “Big 4” (Bosch, Continental, ZF, Denso) surged from 0.58 to **0.83**.
- **Structural Decoupling**: Automakers monopolize end-to-end cognitive algorithms; Tier-1s are commoditized into standard hardware suppliers.

14. Theoretical Contributions & Innovation Policy

1. CPS & Modularity Theory

Extends modularity beyond cognitive inertia (Henderson & Clark, 1990) and mechanical symbiosis (Ulrich, 1995; MacDuffie, 2013). Proves centralized architectures create deterministic design rules (Baldwin & Clark, 2000; Brusoni et al., 2001) for embodied AI.

2. Empirical Methodology

Bridges 768-dim NLP text embeddings with staggered quasi-natural experiments. Solves patent classification lag by capturing micro-level semantic convergence.

3. Firm Vertical Boundaries

Advances property rights theory of architectural control. Proves API standardization triggers downstream assembler invasion and upstream supplier commoditization.

Thank you very much for your time, guidance, and feedback!

Questions, Comments, and Critique are Warmly Welcomed.